

AI Usage Documentation

COP 4610 Project 1 - Shared Memory

Name: Nafiul Khalid

PID: 6436457

Username: nkhal014

In this project, I have utilized artificial intelligence to make informed decisions. The use was intentional and was solely utilized to ensure all the submission requirements are being performed. My AI usage for this assignment are as follow:

1. A Submission TO-DO list:

Using ChatGPT Plus, I have generated a submission TO-DO list. I have attached the given two pdfs:

- COP 4610_Project_1 (Shared memory).pdf
- programming_ass1_help_slide.pdf

With the prompt:

“

Read both the documents, specially the submission requirements, and create a TO-DO list as file ‘checklist.txt’. It should have lists using ‘[]’. Also, add any specific terms or headers or variable names specifically mentioned to use.

“

It generated the checklist.txt file. Then, I read it over and edited the list in my words.

2. A Grading checklist:

Then, in the same conversation chat of (1), I attached the edited ‘checklist.txt’. Then I prompted:

“

Review the attachment. Using the instruction PDFs given by my professor, add a grading checklist under the TO-DO list. The purpose of the grading checklist is to check them when I complete the project to double-check that I am not missing anything for grading.

“

The checklist.txt file was updated. I downloaded the updated file and edited the checklist in my words.

3. Debugging a Shared Memory Key Issue

When I ran the program on the Ocelot server, I received the following error:

“

shmctl: Operation not permitted

“

Then in the same chat, I attached a screenshot of the terminal and prompted ChatGPT:

“

Why the shared memory location is not working.

”

ChatGPT gave me an explanation of how the shared memory key works and suggested me to utilize a personalized space that is “6457” that is unique as it is the last four digit of my PID “6436457”.

The problem was clear when I also ran:

ipcs -m

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nafiul_khalid — ssh nkhal014@ocelot.aul.fiu.edu — 91x58

[nkhal014@ocelot:~/os26 116% ./shared_memory
From Process 2: counter = 214755.
From Process 1: counter = 211751.
From Process 3: counter = 439818.
From Process 4: counter = 525275.
Child with ID: 1525239 has just exited.
Child with ID: 1525240 has just exited.
Child with ID: 1525241 has just exited.
Child with ID: 1525242 has just exited.
shmctl: Operation not permitted
[nkhal014@ocelot:~/os26 117% ipcs -m

----- Shared Memory Segments -----
key      shmid    owner      perms      bytes      nattch     status
0x00000000 294918    jgali028   666        4          0
0x00000000 294919    jgali028   666        4          0
0x00000000 294920    jgali028   666        4          0
0x00000000 294921    jgali028   666        4          0
0x00000000 294922    jgali028   666        4          0
0x00000000 294923    jgali028   666        4          0
0x00000000 294924    jgali028   666        4          0
0x00000000 294925    jgali028   666        4          0
0x00000000 294926    jgali028   666        4          0
0x00000000 262159    dreye134   666        4          0
0x00000000 294928    jgali028   666        4          0
0x00000000 294929    jgali028   666        4          0
0x00000000 294930    jgali028   666        4          0
0x00000000 294931    jgali028   666        4          0
0x00000000 294932    jgali028   666        4          0
0x00000000 294933    jgali028   666        4          0
0x00000000 294934    jgali028   666        4          0
0x00000000 294935    jgali028   666        4          0
0x00000000 294936    jgali028   666        4          0
0x00000000 294937    jgali028   666        4          0
0x00000000 294938    jgali028   666        4          0
0x00000000 294939    jgali028   666        4          0
0x00000000 294940    jgali028   666        4          0
0x00000000 294941    jgali028   666        4          0
0x00000000 294942    jgali028   666        4          0
0x00000000 294943    jgali028   666        4          0
0x00000000 294944    jgali028   666        4          0
0x00000000 294945    jgali028   666        4          0
0x00000000 294946    jgali028   666        4          0
0x00000000 294947    jgali028   666        4          0
0x00000000 294948    jgali028   666        4          0
0x00000000 294949    jgali028   666        4          0
0x00000000 294950    jgali028   666        4          0
0x00000000 294951    jgali028   666        4          0
0x00000000 294952    jgali028   666        4          0
0x00000000 294953    jgali028   666        4          0
0x00000000 294956    dvall059   600        4          0
0x00000000 229425    dreye134   666        4          0
0x000005d9 819250    amart1432   666        4          0

[nkhal014@ocelot:~/os26 118% nano sh
shared_memory*  shared_memory.c

```

9. Conclusion

AI was used as a support tool during this project to help me understand the assignment requirements, troubleshoot an error, organize the program results, and prepare the required submission files.